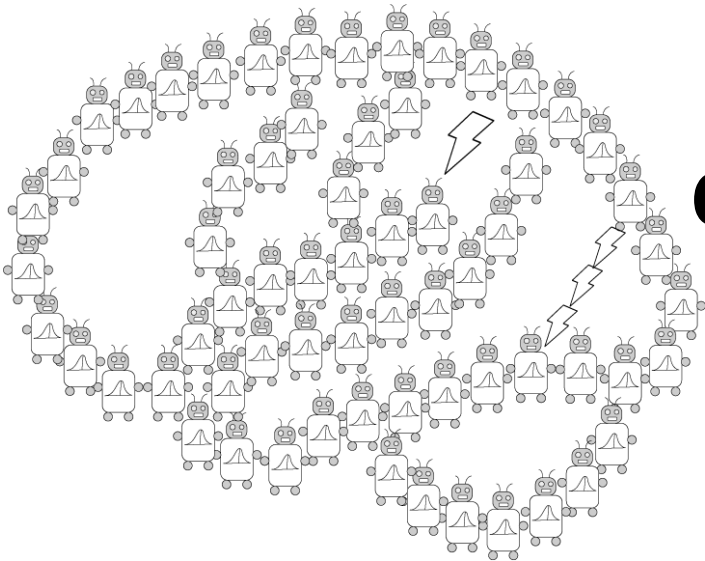


(Causal) Bayesian Networks



**PSYCH/COMP SCI/BMI 841:
Computational Cognitive Science**

Joseph Austerweil

February 27, 2025

- Admin Matters
- Bayesian Networks
- Causal Bayesian Networks
- Introduction to Information Theory
- **Short rest 1**
- Student presentations:
 - Elijah presenting Lagnado & Sloman (2002)
 - Yinsu presenting Tenenbaum & Griffiths (2001)

Administrative Matters

- Class ends about 250-3pm (for Joe)
- Discussion posts will be a bit delayed this week.
- Generalization Assignment due Mar 5 8:59pm
 - That's next Wednesday at 8:59pm!
 - Remember you have three “free” late days for assignments.
 - Late days do not prorate (used up in 24-hour chunks).
- Project proposal due Mar 12 8:59pm
 - No late days for project proposal (or final paper/presentation)
- Questions?

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The Markov assumption

Every node is conditionally independent of its non-descendants, given its parents

$$P(x_i \mid x_{i+1}, \dots, x_k) = P(x_i \mid \mathbf{Pa}(X_i))$$

where $\mathbf{Pa}(X_i)$ is the set of parents of X_i

$$P(x_1, \dots, x_k) = \prod_{i=1}^k P(x_i \mid \mathbf{Pa}(X_i))$$

(via the product rule)

The two representations are equivalent via Hammerly-Clifford's Theorem...

Reading a Bayesian network

The structure of a Bayes net can be read as the *generative process* behind a distribution.

Gives the joint probability distribution over variables obtained by sampling each variable conditioned on its parents.

Simple rules for determining whether two variables are dependent or independent.

Reading a Bayesian network

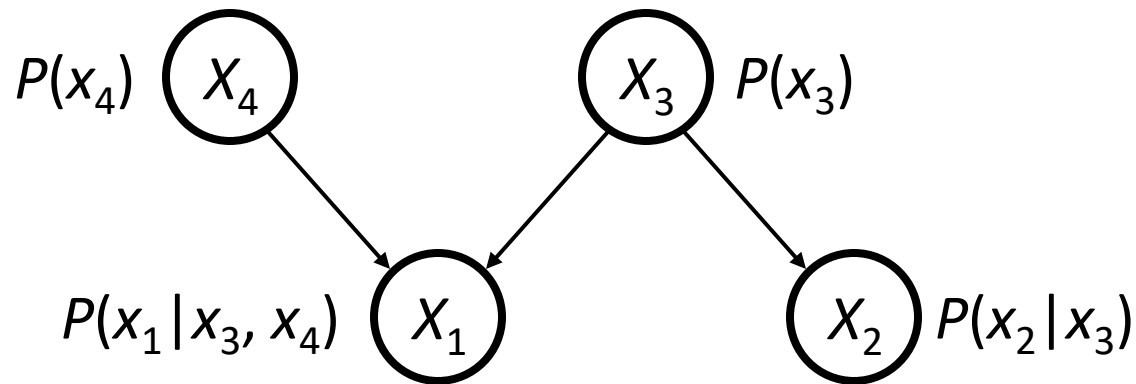
Four random variables:

X_1 coin toss produces heads

X_2 pencil levitates

X_3 friend has psychic powers

X_4 friend has two-headed coin



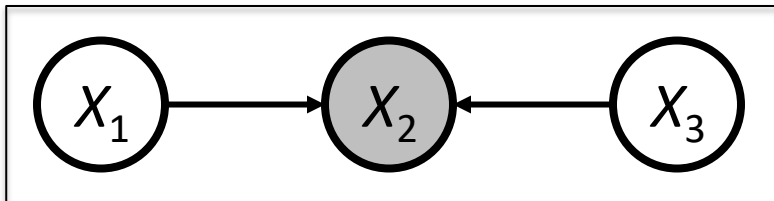
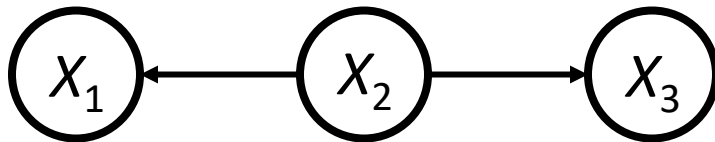
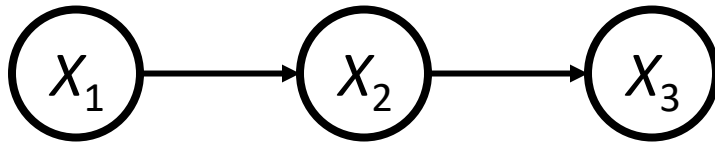


Identifying independence

Can identify independence relations for any graph with the rules for all Directed Acyclic Graphs of 3 variables (obs and unobs).

E.g., X_1 is sprinklers were on, X_2 is grass is wet, X_3 it rained...

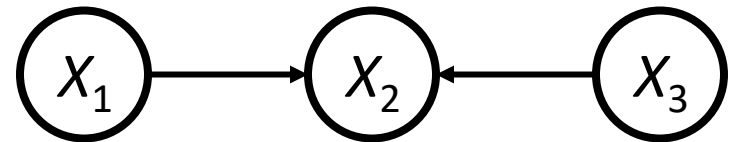
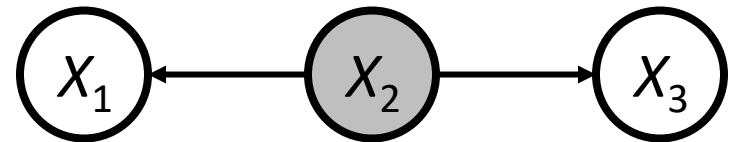
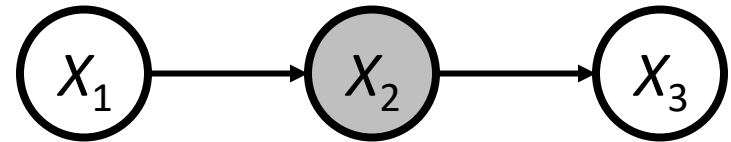
X_1 and X_3 dependent



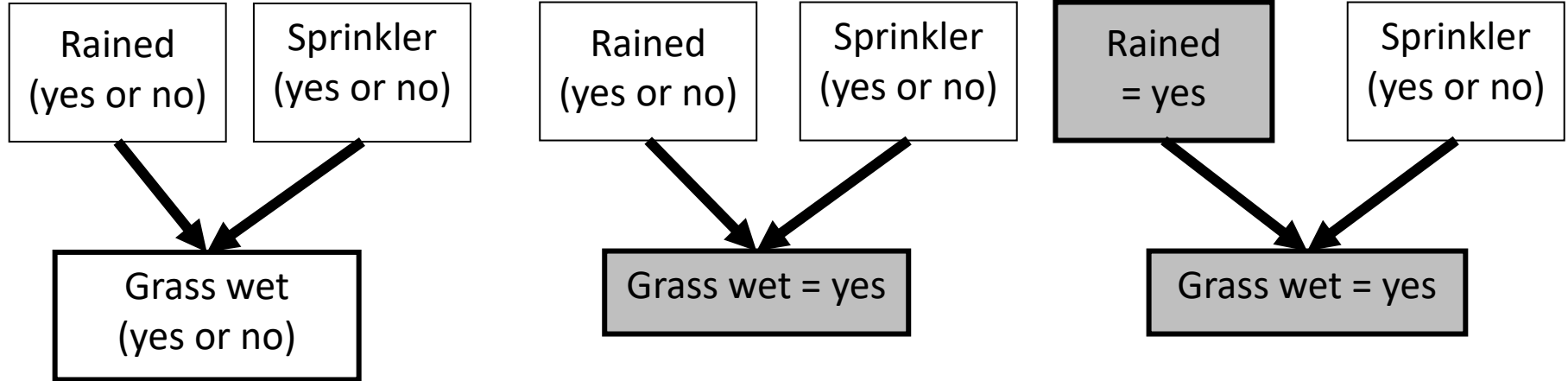
Why?

(shaded variables are observed)

X_1 and X_3 independent



Explaining Away



Corresponding probability relations

Unconditional independence:

$$P(\text{Sprinkler} = \text{yes} \mid \text{Rained} = \text{yes}) = P(\text{Sprinkler} = \text{yes} \mid \text{Rained} = \text{no})$$

After learning effect:

$$P(\text{Sprinkler} = \text{yes} \mid \text{Grass wet} = \text{yes}) > P(\text{Sprinkler} = \text{yes})$$

After learning effect and one cause:

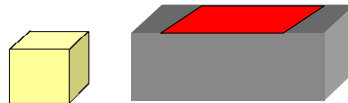
$$\begin{aligned} P(\text{Sprinkler} = \text{yes} \mid \text{Grass wet} = \text{yes}, \text{Rained} = \text{yes}) &= P(\text{Sprinkler} = \text{yes}) \\ &< P(\text{Sprinkler} = \text{yes} \mid \text{Grass wet} = \text{yes}) \end{aligned}$$

Blicket detector

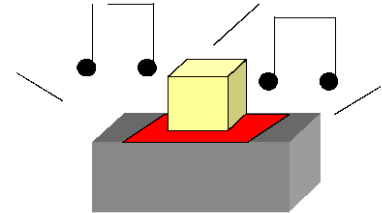
(Dave Sobel, Alison Gopnik, and colleagues)



See this? It's a
blinket machine.
Blinkets make it go.



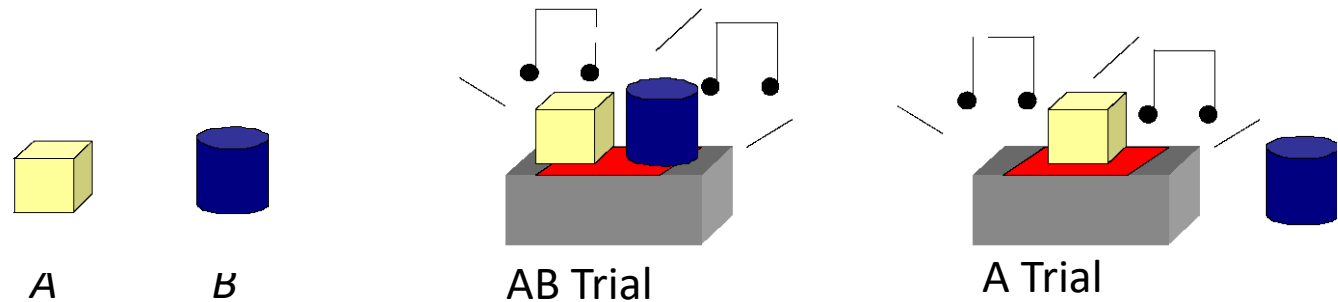
Let's put this one
on the machine.



Oooh, it's a
blinket!

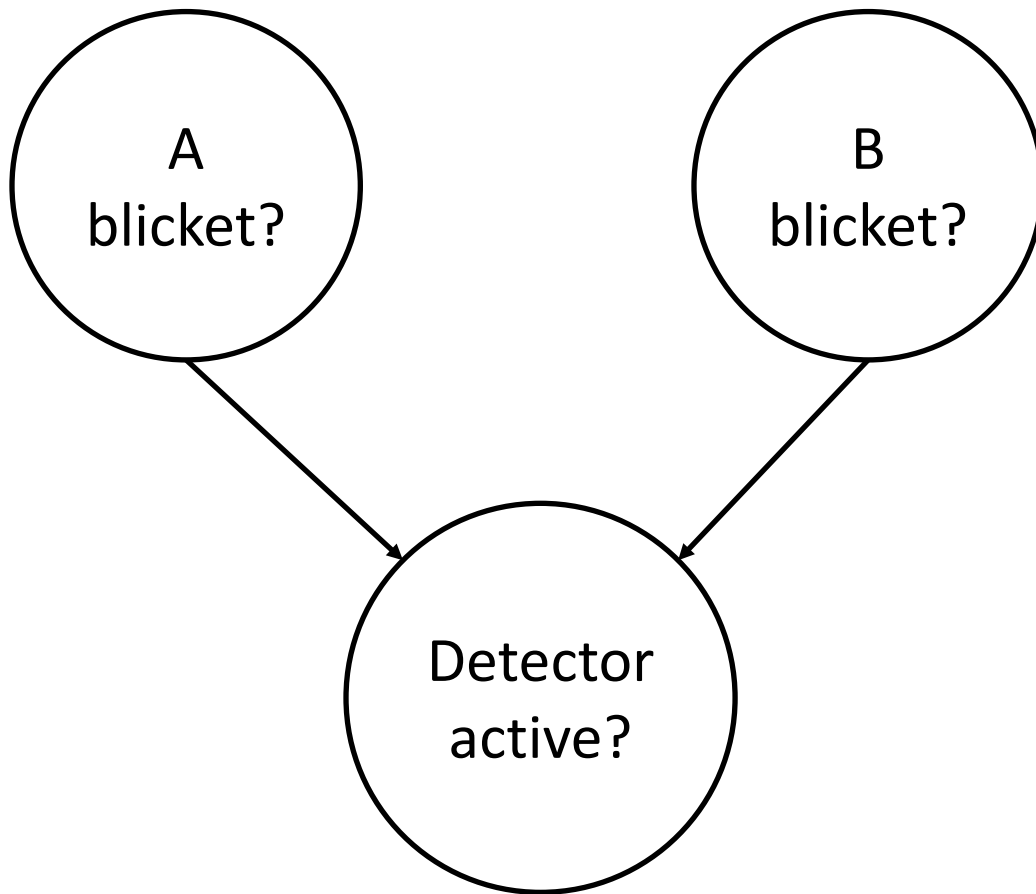
“Backwards blocking”

(Sobel, Tenenbaum & Gopnik, 2004)



- Two objects: A and B
- Trial 1: A B on detector – detector **active**
- Trial 2: A on detector – detector **active**
- 4-year-olds judge whether each object is a blicket
 - A: a blicket (100% say yes)
 - B: probably not a blicket (34% say yes)

Bayes net for backwards blocking

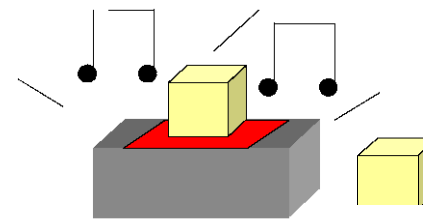
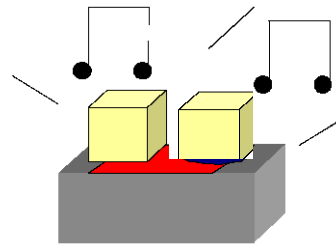
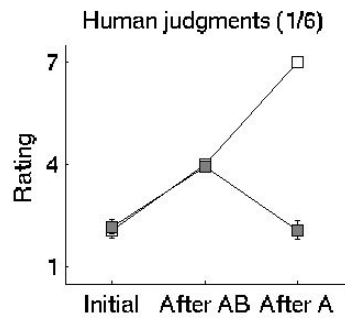
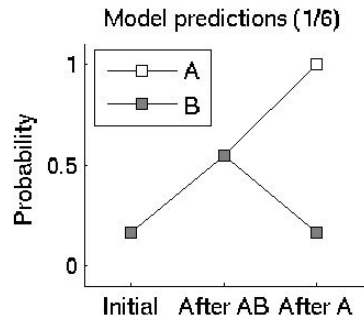


The “causes” for the detector activating are whatever objects are put on it that trial.

Minor note:
Slightly different Bayes nets between trials.

Manipulating prior probability

(Griffiths, Sobel, Tenenbaum, & Gopnik, 2011)



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Causality

Asymmetric relation between variables:

A cause brings about an effect via a mechanism.

Defining these terms is very very tricky...

But, it is critical, pervasive, and domain-general:

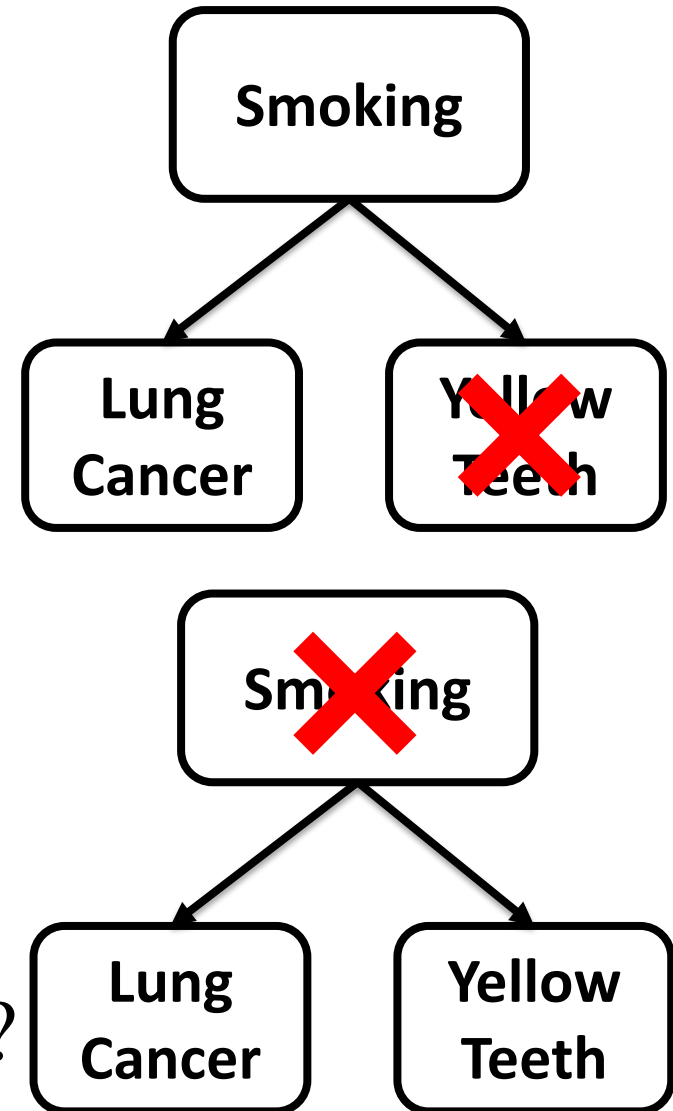
- The cue ball causes the 8-ball to move
- Hot stoves cause burned fingers
- Incitement causes violence
- Slavery caused the civil war
- Smoking causes lung cancer
- Genes cause eye color

A quick word on causality...

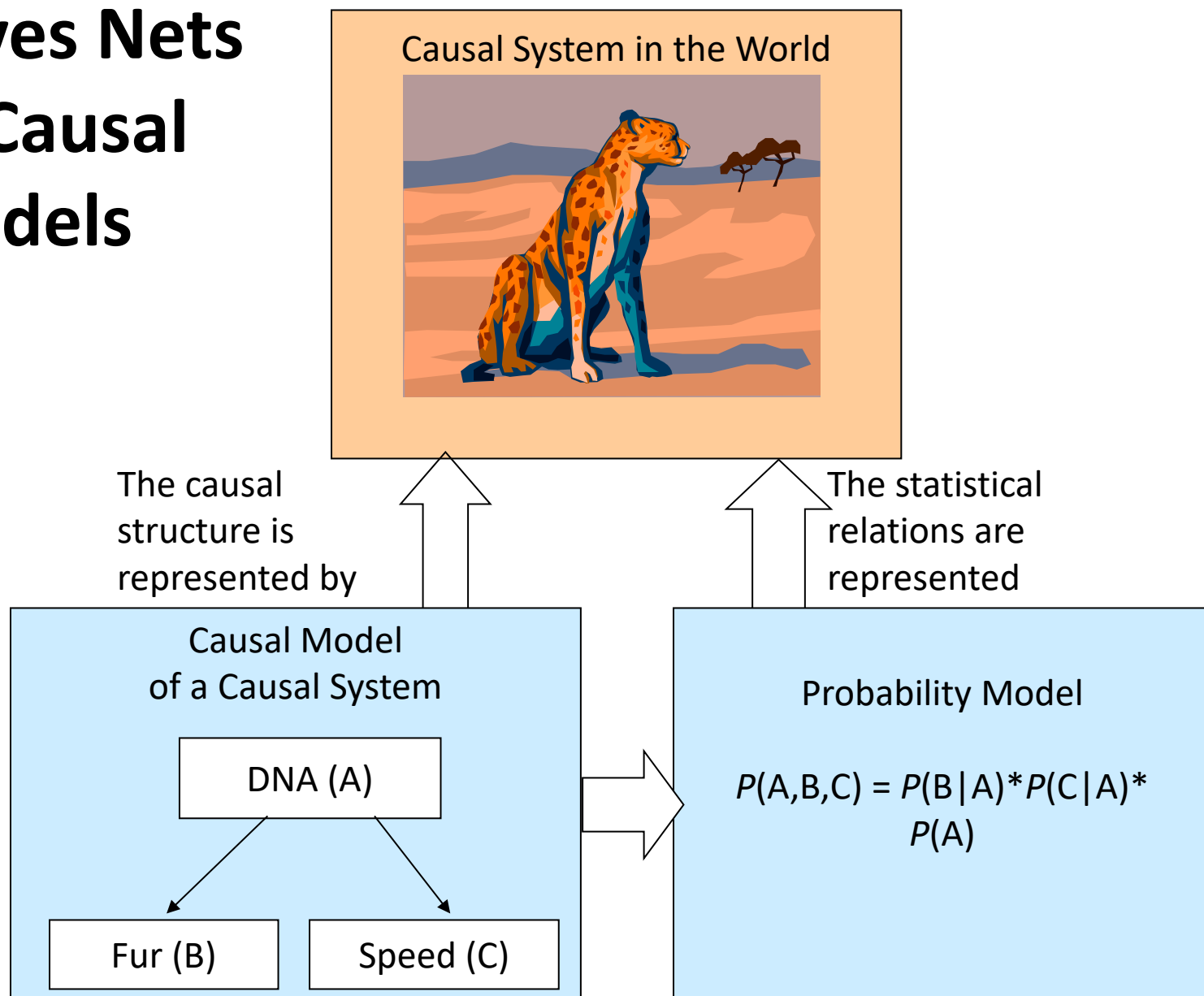
An evidential relation: Having yellow teeth is correlated with lung cancer.

Therefore, you should whiten your teeth to lower the probability of getting lung cancer.

What's wrong with this argument?



Bayes Nets as Causal Models

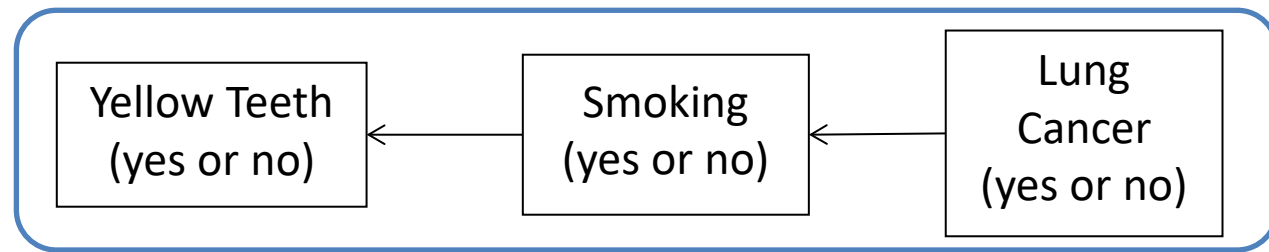


(Pearl, 1988, 2000; Spirtes, Glymour, & Scheines, 1993, Sloman, 2005)



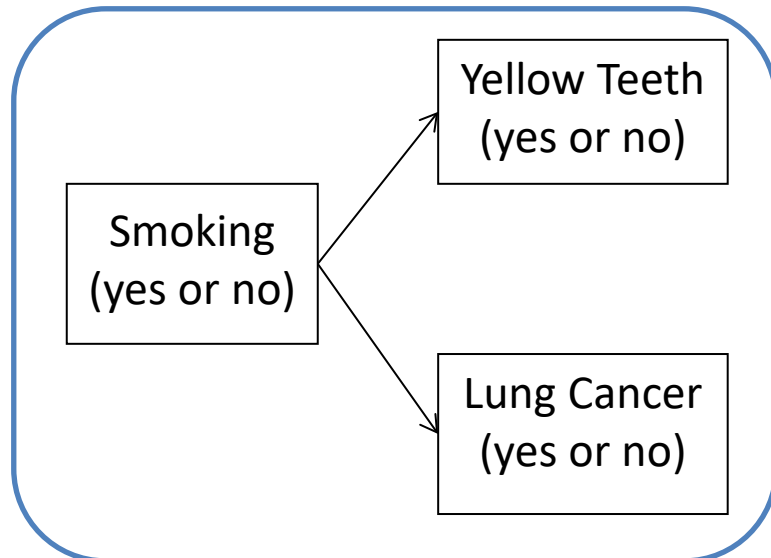
Limits of causation from mere observation: Markov equivalence classes

Get evidence of pairwise correlations between smoking, yellow teeth, and lung cancer. What can you infer?



$$P(Y, S, L) = P(L)P(S|L)P(Y|S)$$

$$\uparrow$$
$$P(L|S)P(S)$$

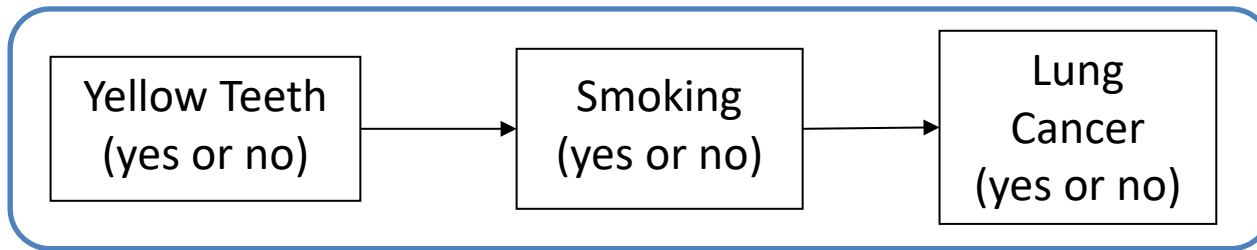


$$P(Y, S, L) = P(S)P(Y|S)P(L|S)$$



Limits of causation from mere observation: Markov equivalence classes

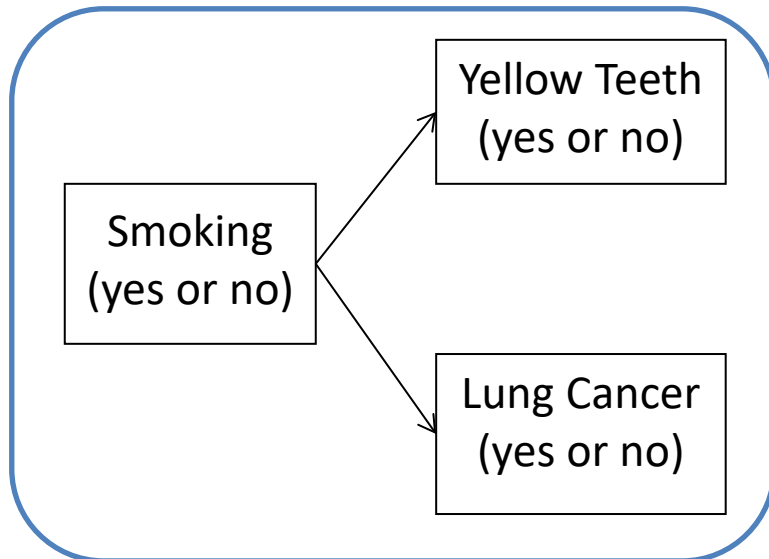
Get evidence of pairwise correlations between smoking, yellow teeth, and lung cancer. What can you infer?



$$P(Y, S, L) = \boxed{P(Y)P(S|Y)}P(L|S)$$

↓

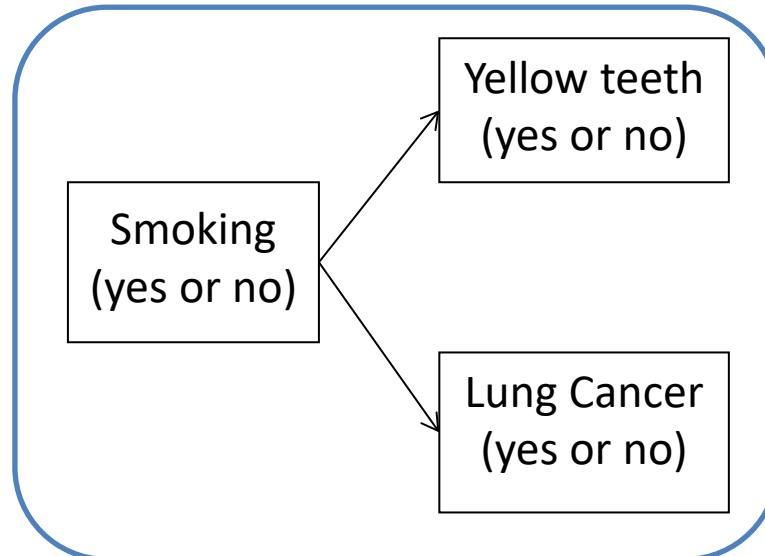
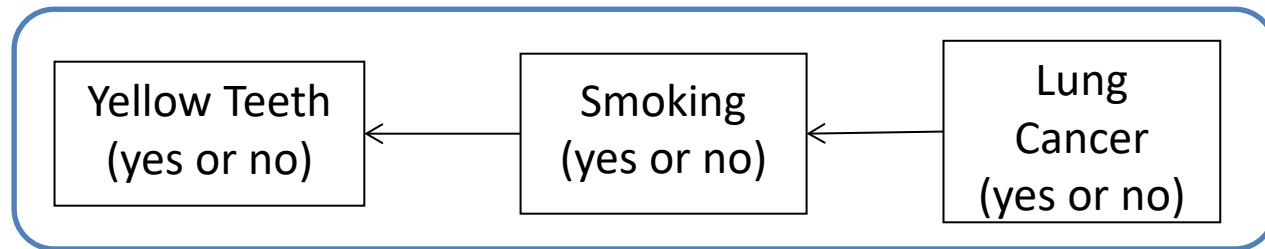
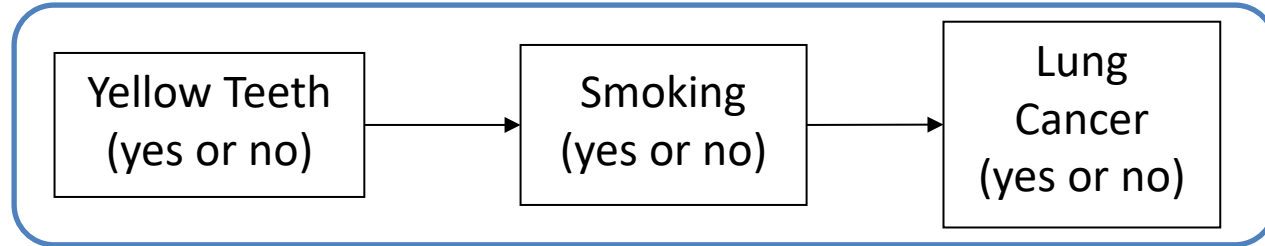
$$\boxed{P(S)P(Y|S)}$$



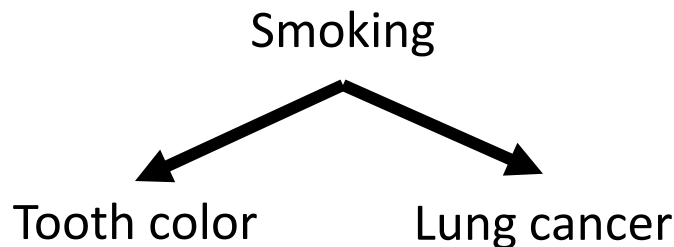
$$P(Y, S, L) = P(S)P(Y|S)P(L|S)$$

Limits of causation from mere observation:

Markov equivalence classes

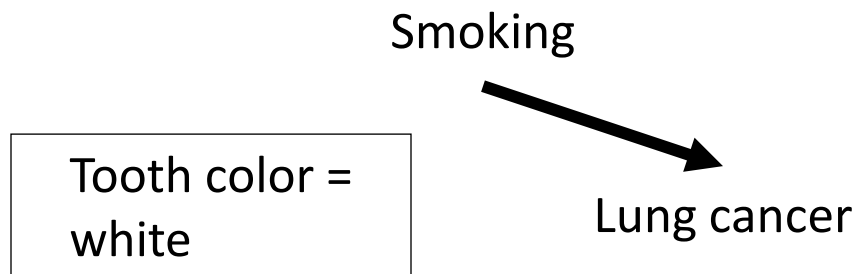


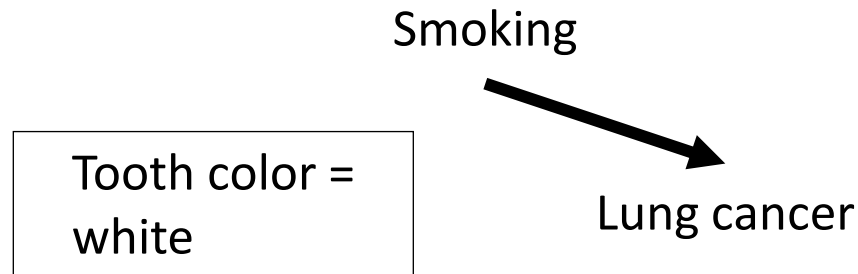
The Logic of “Doing”



To calculate the causal effect of whitening teeth on lung cancer, first *do*(Tooth color = white):

1. Set Tooth color to white.
2. Eliminate the causal links to it:





We see that Lung cancer is unaffected by Tooth color, so

$$\Pr\{\text{Lung cancer} = \text{yes} \mid \underline{do(\text{Tooth color} = \text{white})}\} \\ = \Pr\{\text{Lung cancer} = \text{yes} \}$$

Pearl (2000) calls this a *surgery*.

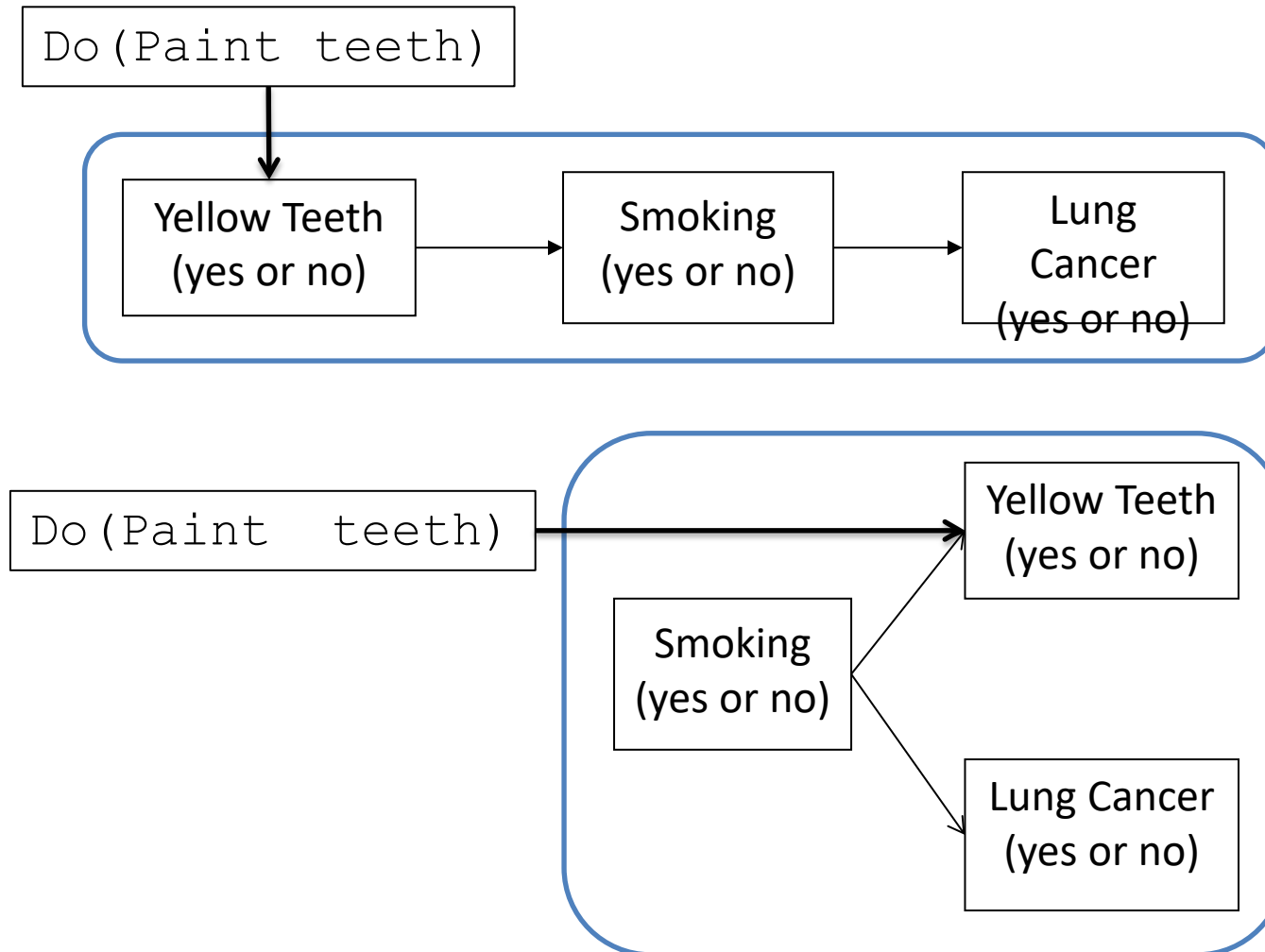
People represent *autonomous mechanisms* so that the effect of interventions can be isolated.

A causal model is an *oracle for interventions*.

Now can distinguish between chains and common cause!

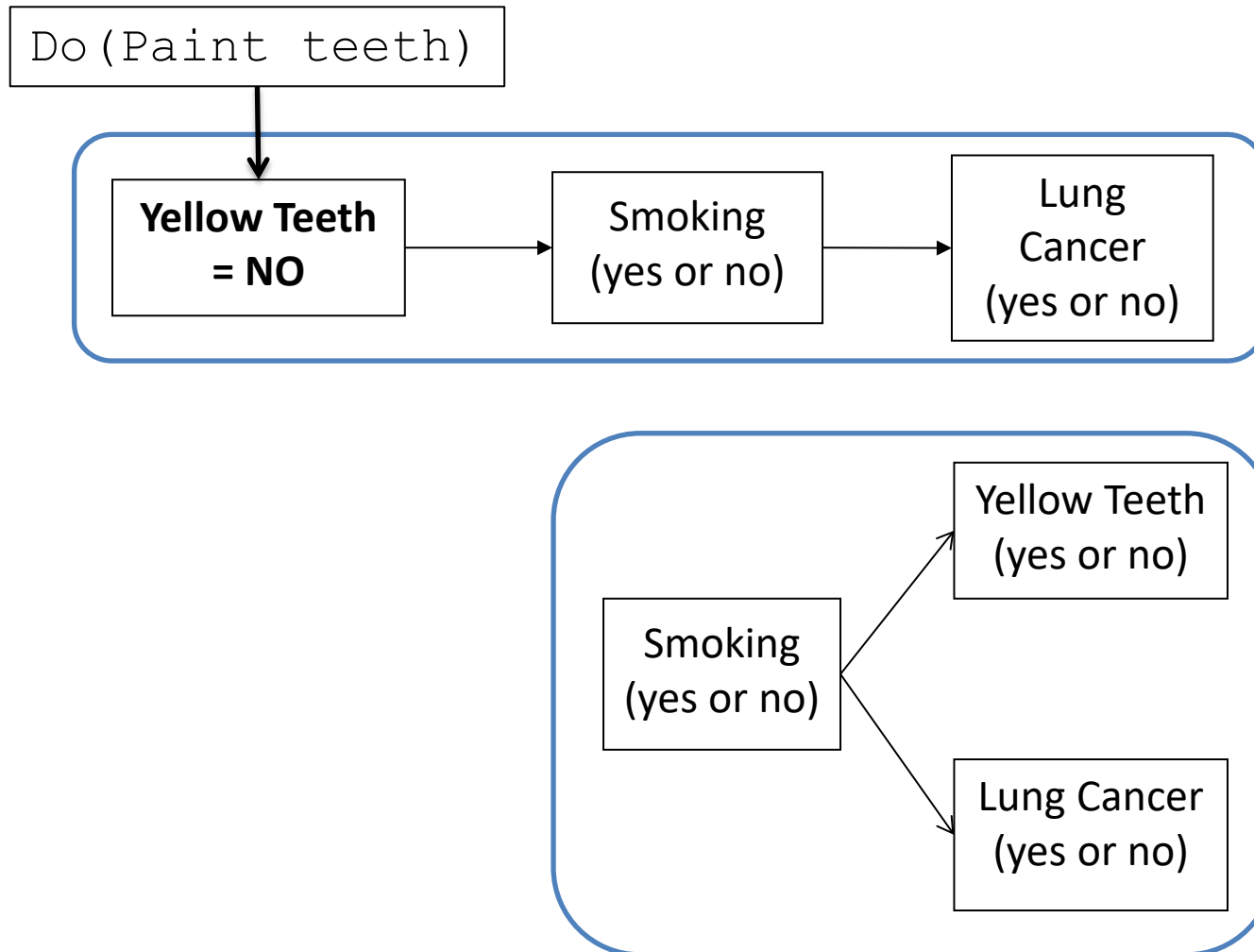
Limits of causation from mere observation: Markov equivalence classes

How do you intervene to distinguish between chain and common cause?



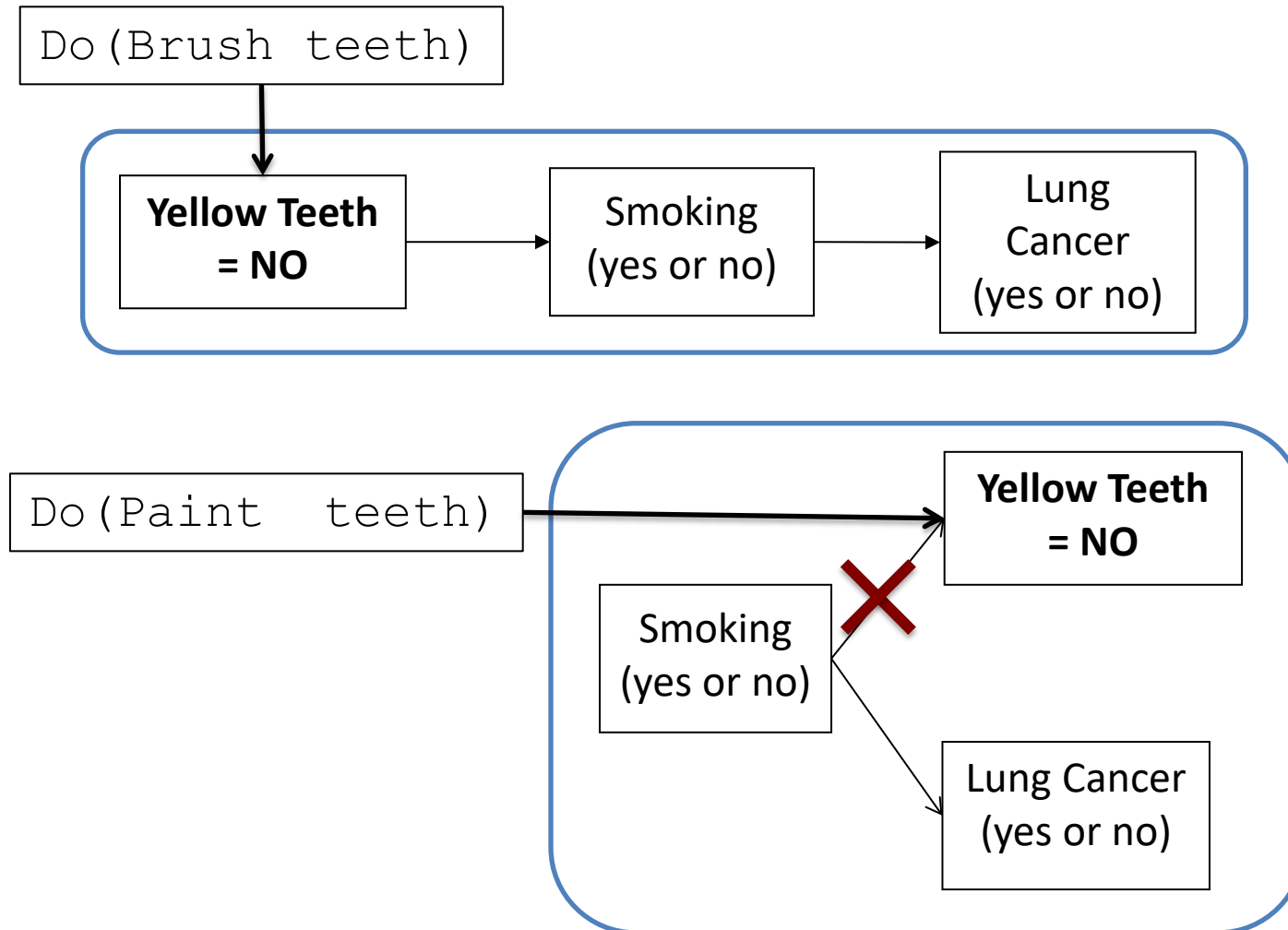
Limits of causation from mere observation: Markov equivalence classes

How do you intervene to distinguish between chain and common cause?



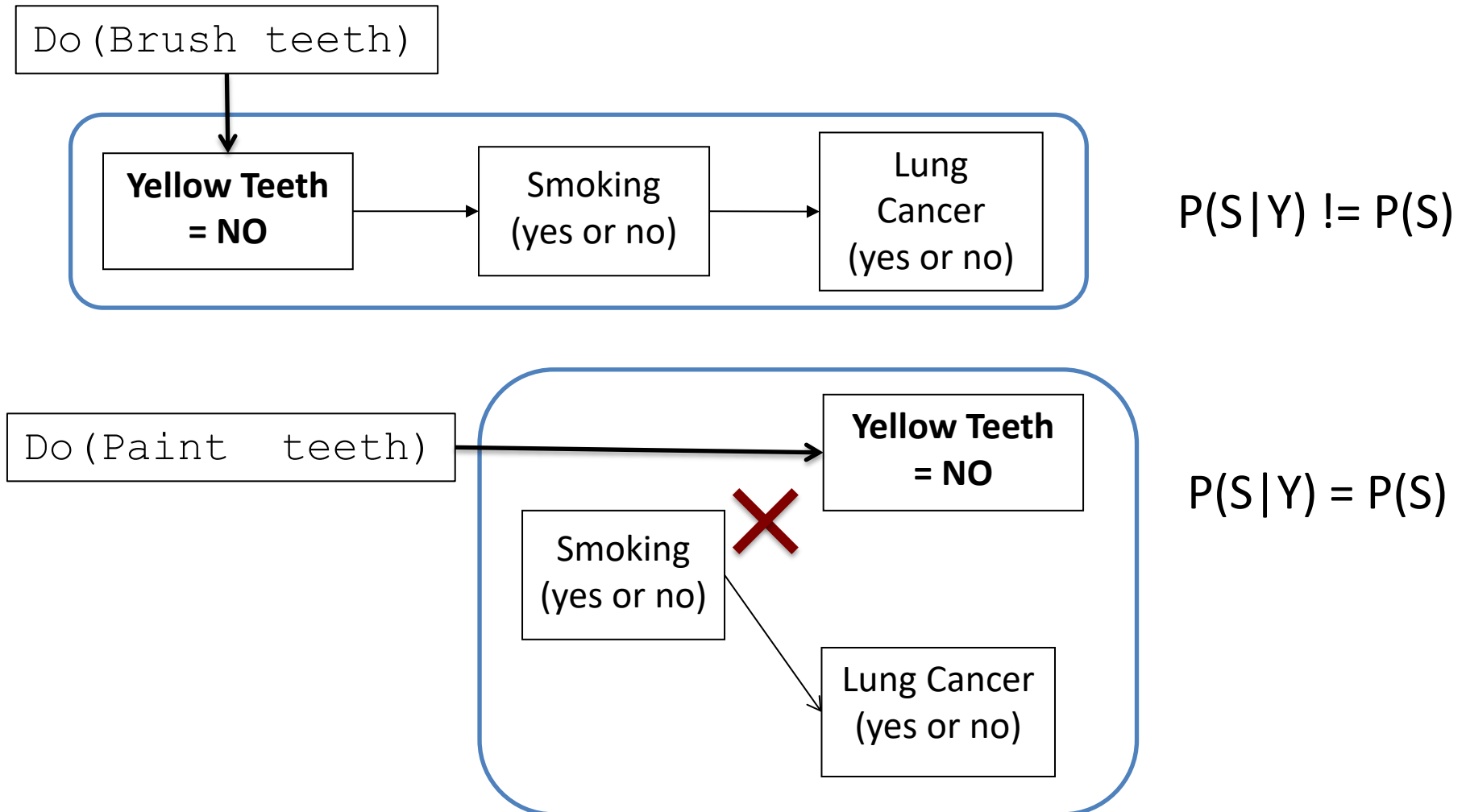
Limits of causation from mere observation: Markov equivalence classes

How do you intervene to distinguish between chain and common cause?



Limits of causation from mere observation: Markov equivalence classes

How do you intervene to distinguish between chain and common cause?



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Uncertainty, Randomness, and Information

Probability takes uncertainty as a basic concept.

How much information do we gain from observing a random event?

Most common measure is Shannon's Entropy

$$H(\vec{p}) = E[-\log \vec{p}] = - \sum_k p_k \log p_k$$

Remember:

$$E[f(X)] = \sum_x P(X = x) f(x)$$

Why Shannon's Entropy

Follows via an inevitability proof...

If the following things are true:

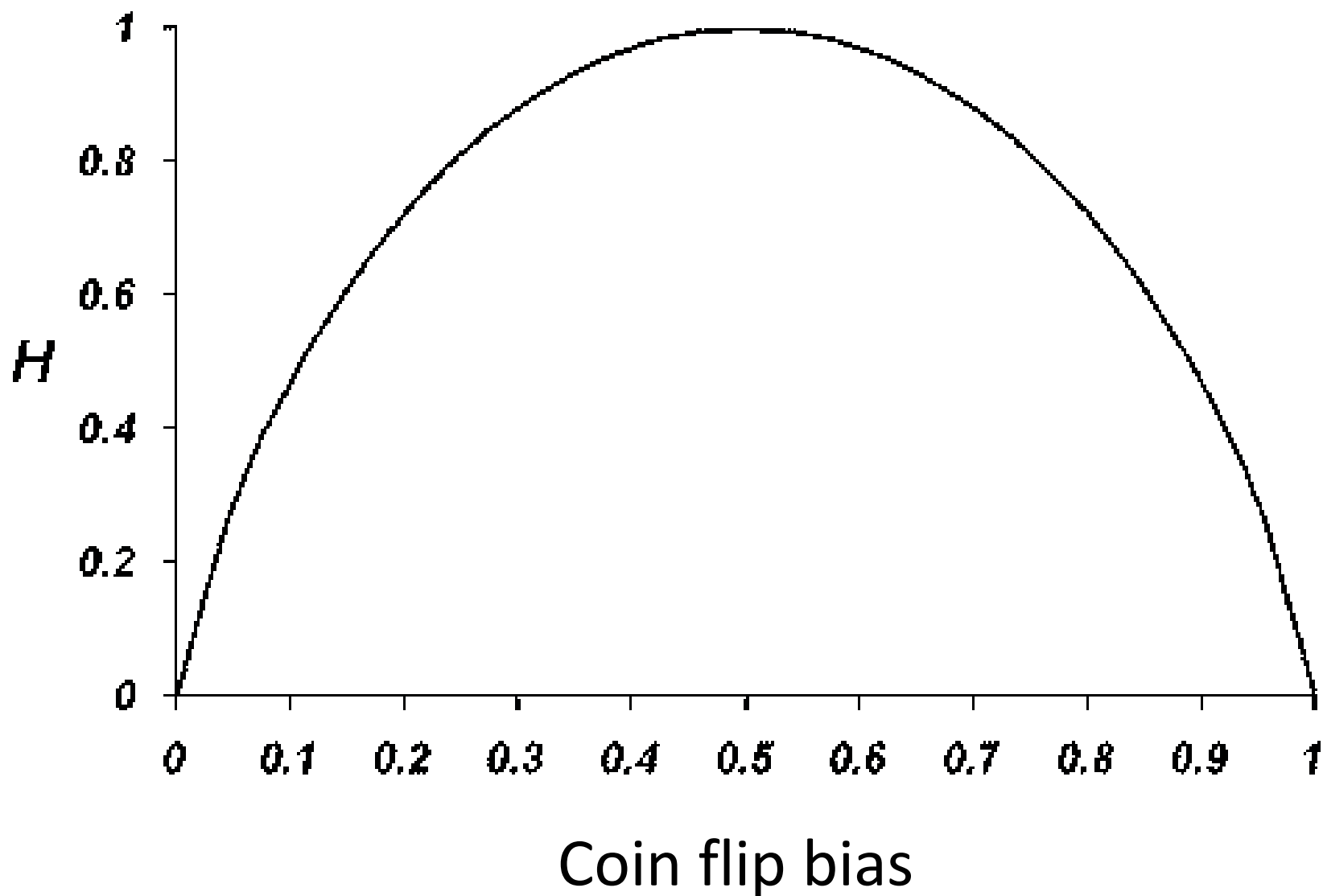
- (1) measure of information is non-negative
- (2) Learning the outcome of a deterministic event provides no information
- (3) Information of independent events can be “split”:

$$I(p_1, p_2) = I(p_1) + I(p_2)$$

⇒ Shannon proved only $H(x)$ satisfies these three properties.

⇒ Note: Shannon's Entropy isn't the only one used and there is really interesting research on a broader framework called the “Sharmal-Mittal Framework”

Shannon's Entropy for Coinflip



Joint Entropy and Conditional Entropy

Joint and Conditional Entropy are

$$H(X, Y) = E[-\log P(x, y)] = - \sum_{x \in X, y \in Y} P(x, y) \log P(x, y)$$

$$H(X | Y) = E[H(X | Y = y)] = - \sum_{x \in X, y \in Y} P(x, y) \log P(x | Y = y)$$

$$H(X | Y) = \sum_{y \in Y} P(Y = y) \sum_{x \in X} P(X = x | Y = y) \log P(X = x | Y = y)$$

Note: $H(X|Y) = H(X, Y) - H(Y)$

$$\begin{aligned} & - \sum_{x,y} P(x, y) \log P(x, y) - \sum_y -P(y) \log P(y) \\ & - \sum_{x,y} P(x, y) \log P(x, y) - \sum_y P(y) \log \frac{1}{P(y)} && P(x|y) \\ & - \sum_{x,y} P(x, y) \log P(x, y) - \sum_{x,y} P(x, y) \log \frac{1}{P(y)} \\ & - \sum_{x,y} P(x, y) \log P(x, y) - P(x, y) \log \frac{1}{P(y)} = - \sum_{x,y} P(x, y) \log \frac{P(x,y)}{P(y)} \end{aligned}$$

KL-Divergence

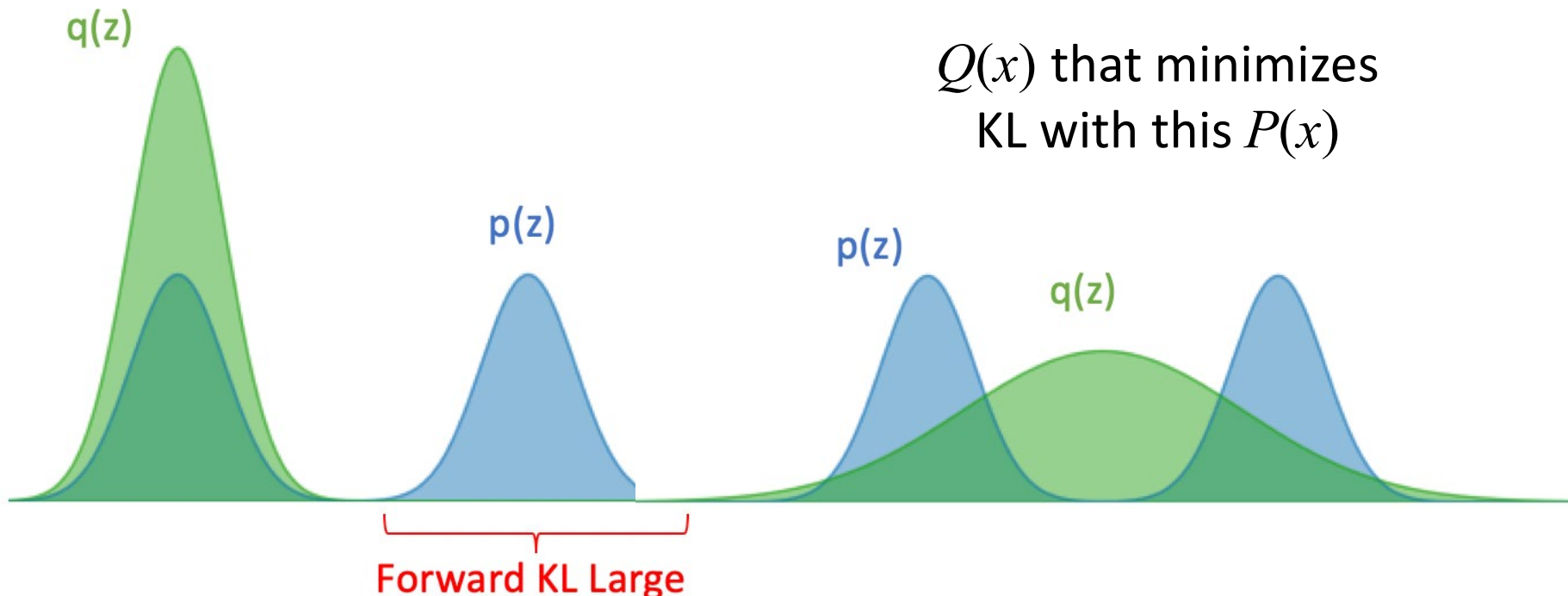
KL-Divergence of distribution $P(X)$ relative to some “proxy” $Q(X)$ is written as:

$$D_{KL}(P(X) \parallel Q(x)) =$$

KL-Divergence is **NOT** a distance metric, but is similar to one.

It is not symmetric and it violates the triangle inequality.

“Forward KL-Divergence”



KL-Divergence

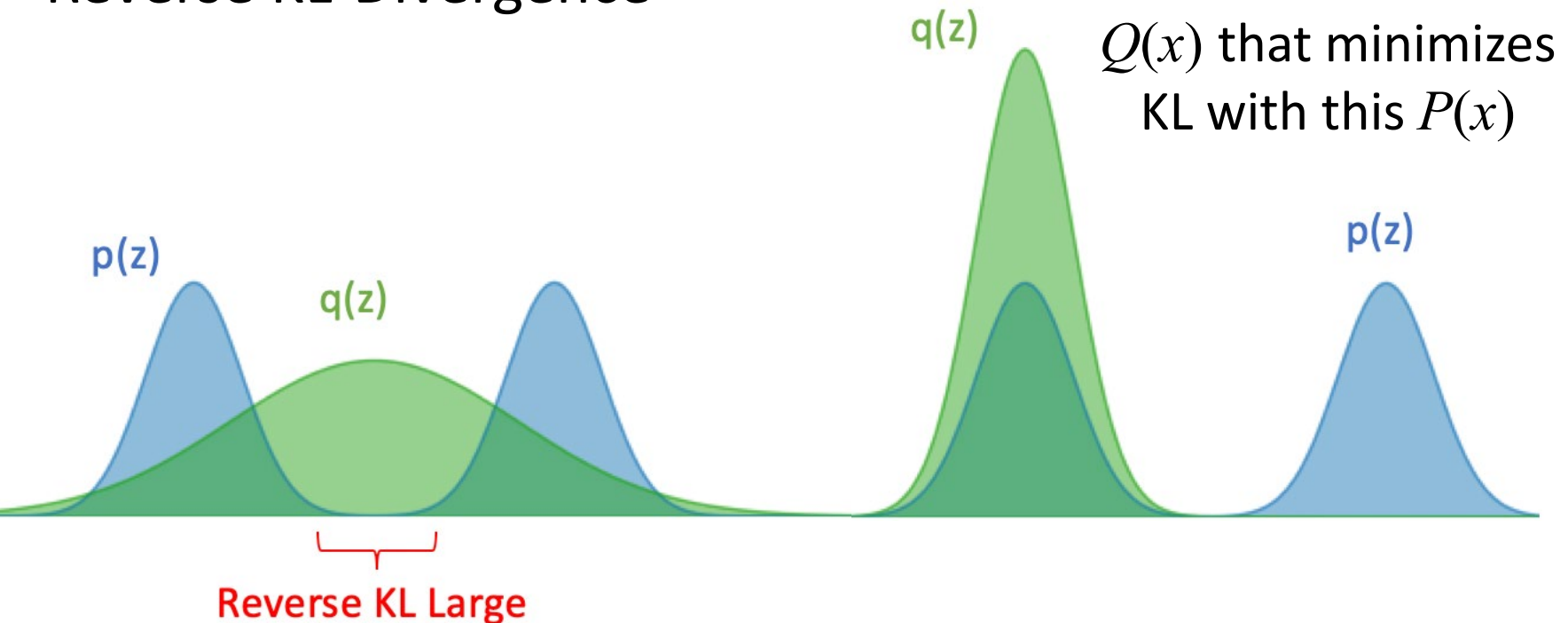
KL-Divergence of distribution $P(X)$ relative to some “proxy” $Q(X)$ is written as:

$$D_{KL}(Q(X) \parallel P(x)) = \sum_x Q(x) \log \frac{Q(x)}{P(x)}$$

KL-Divergence is **NOT** a distance metric, but is similar to one.

It is not symmetric and it violates the triangle inequality.

“Reverse KL-Divergence”



Mutual Information

Mutual information of X and Y is the entropy of their joint distribution as compared to their marginal distributions.

$$I(X; Y) = \sum_{x,y} P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right)$$

Like KL-Divergence, it is non-negative and is large when the joint distribution is similar to the product of the marginals.

Unlike KL-Divergence, it is symmetric.

$$I(X; Y) = H(Y) - H(Y|X) = H(X) - H(X|Y)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = H(X, Y) - H(X|Y) - H(Y|X)$$

To make it a *true* metric, it often is used to define “variation of information”:

$$H(X, Y) - I(X; Y)$$

Cross Entropy

The cross-entropy of using a “proxy” Q to stand in for a “true” distribution P .

It is the average amount of information needed to identify an event from source P sent through a noisy channel, resulting in observed distribution Q .

$$E^P[-\log Q(x)] = - \sum_x P(x) \log Q(x) = H(P(x)) + D_{KL}(P \parallel Q)$$

This is often used as a “loss function” in machine learning.

A loss function is a function describing the amount of error for using an estimated quantity rather than the true one.

$$\text{E.g., } L(x, \hat{x}) = (x - \hat{x})^2 \qquad L(x, \hat{x}) = |x - \hat{x}|$$

Lossless and Lossy Compression

Lossless examples: “zip”, “rar”, “flac”

Lossy examples: “jpg”, “mp3”, “mpeg”

Source coding theorem for Lossless Compression:

N outcomes from source X can be compressed without error to $N H(X)$.

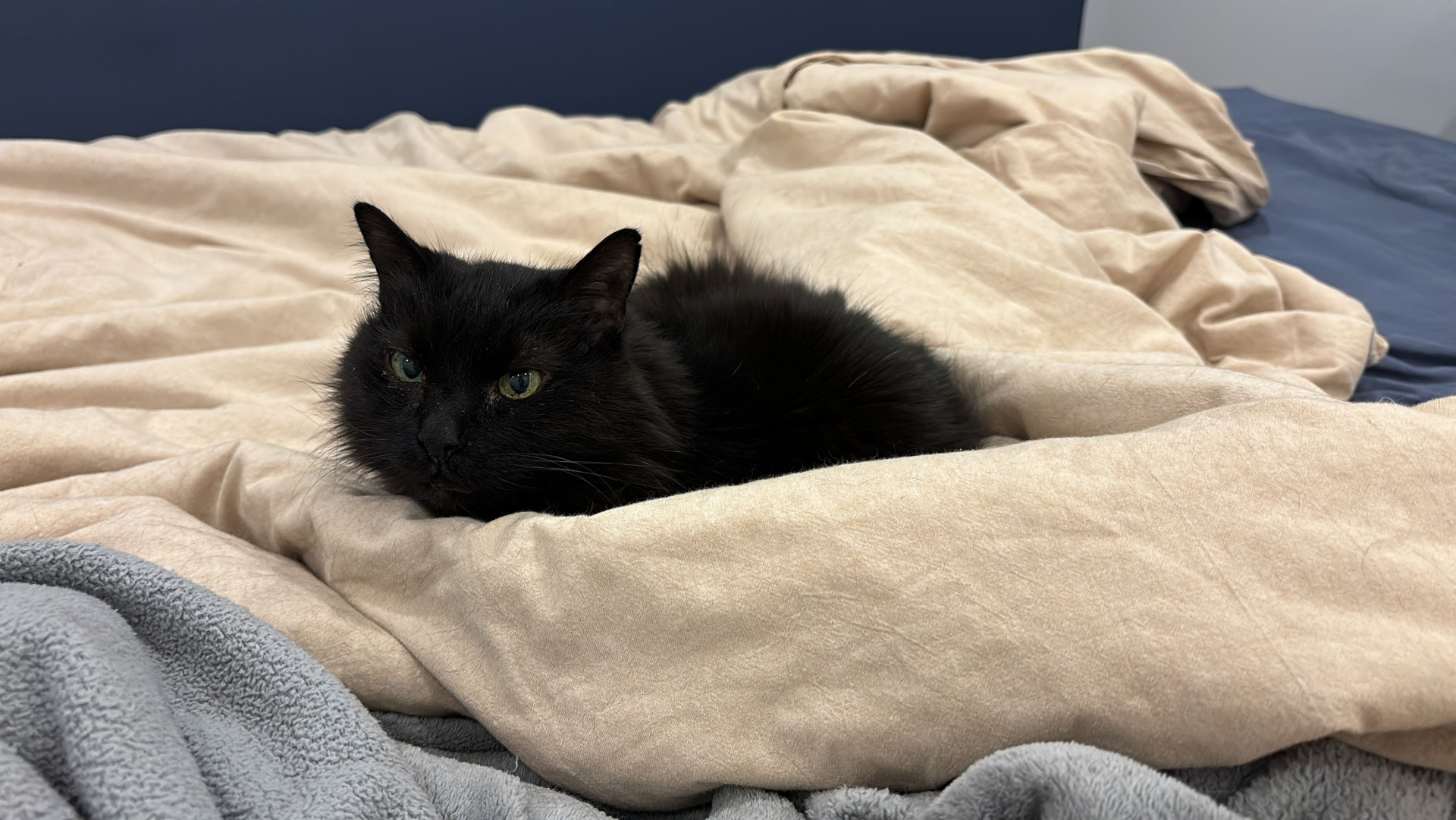
Optimal coding is straightforward and (expected) length per bit is $H(X) < L(X) < H(X) + 1$.

Rate Distortion Theory is the field for lossy compression.

$$\inf_{Q_{Y|X}(y|x)} E[D_Q[X, Y]] \text{ subject to } I_Q(Y; X) \leq R.$$

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